

PRICE SENTINEL: ALL TRACKS RESOURCES GUIDE

Complete Dataset & Tool Reference for All 5 Tracks

TRACK 1: DATA COLLECTION & ENGINEERING

PRIMARY DATASET (Week 1 Priority - Share with Everyone)

Statistics Canada Table 18-10-0245-01: Monthly Average Retail Prices

- **Download:** <https://www150.statcan.gc.ca/n1/tbl/csv/18100245-eng.zip>
- **Interactive:** <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1810024501>
- **What:** Vancouver grocery prices 2017-2025 (milk, eggs, bread, etc.)
- **Format:** CSV, ready to import to Excel/Google Sheets
- **Task:** Clean, filter to Vancouver, share with all tracks by end of Week 1

SUPPLEMENTARY DATASETS

Food Banks Canada - HungerCount 2024

- <https://foodbankscanada.ca/hunger-in-canada/hungercount/>
- **Data:** 2M+ monthly visits, demographics, employment status
- **Key stats:** 18% employed, 32% newcomers, 33% children

Statistics Canada Food Insecurity (Table 13-10-0835-01)

- <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1310083501>
- **BC rate:** 24.4% food insecure

City of Vancouver Census Income Data

- <https://opendata.vancouver.ca/explore/dataset/census-local-area-profiles-2016/>
- **Format:** CSV with income by neighborhood

BC Minimum Wage History

- <https://www2.gov.bc.ca/gov/content/employment-business/employment-standards-advice>
- **Use:** Compare wage growth vs. food price inflation

TOOLS

Google Sheets (recommended):

- Free, collaborative, no coding needed
- Share link with "anyone can view"

Airtable:

- <https://airtable.com> (free tier)
- Advanced database features

Excel

- Good for offline work
- Powerful formulas and pivot tables
- Familiar to most users

DATA COLLECTION METHODS

Option A: Use Existing Datasets (Easiest)

- Download StatsCan CSV
- Compile publicly available price reports
- Use government open data portals

Option B: Manual Research

- Visit grocery store websites
- Copy prices into spreadsheet
- Document source URL and date
- Takes 2-3 hours for 5-10 stores

Option C: Automated Data Collection (Advanced)

- Use Python scripts to collect data from online sources
- Requires programming knowledge
- Only if you're comfortable with coding

DELIVERABLE

Clean CSV with 50+ stores, 10-15 items, StatsCan baseline + optional scraped prices



TRACK 2: RESEARCH & DATA ANALYSIS

YOUR DATA SOURCES

From Track 1 (Week 1):

- Cleaned StatsCan Vancouver prices CSV

Direct Statistical Analysis:

Consumer Price Index - Food (Table 18-10-0004-03)

- <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1810000403>
- **What:** Month-over-month food inflation rates
- **Use:** Calculate "eggs up 70%, wages up 28%" comparisons

Labour Force Survey (Table 14-10-0287-01)

- <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1410028701>
- **What:** Employment, wages, unemployment by region
- **Use:** Income context for affordability analysis

Low-Income Statistics (Table 11-10-0135-01)

- <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1110013501>
- **What:** Poverty rates, low-income cut-offs
- **Use:** Define "low-income household" thresholds

CMHC Rental Market Data

- <https://www03.cmhc-schl.gc.ca/hmip-pimh>
- **What:** Average rents Metro Vancouver
- **Use:** Housing cost context

TOOLS

Excel/Google Sheets:

- Pivot tables for price comparisons
- Formulas: =AVERAGE(), =PERCENTILE(), =IF()
- Charts: line graphs for trends, bar charts for comparisons

Free Data Visualization:

- **Datawrapper:** <https://www.datawrapper.de>
- **Flourish:** <https://flourish.studio>
- **Google Charts:** <https://developers.google.com/chart>

Statistical Analysis:

- **Python Pandas:** <https://pandas.pydata.org>
- **R (free):** <https://www.r-project.org>

KEY ANALYSES

1. **Affordability Index**
 - a. Calculate Vancouver Essential Basket avg price
 - b. Rank stores: % above/below average
 - c. Color-code: Green (<-15%), Yellow (±10%), Red (>+10%)
2. **Household Savings Calculator**
 - a. Input: Family size, current store, potential store
 - b. Output: Monthly savings, annual savings
 - c. Formula: (Store A basket - Store B basket) x 4.3 weeks/month
3. **Historical Trend Analysis**
 - a. 2017 vs 2025 price comparison
 - b. Inflation rate by item category
 - c. Wage growth vs food price growth
4. **Geographic Analysis**
 - a. Map neighborhoods by store density
 - b. Identify "food deserts" (no affordable store within 2km)
 - c. Overlay with Census low-income data

DELIVERABLE

10-15 page PDF report with: Affordability Index table, savings calculator, trend charts, geographic analysis



TRACK 3: DESIGN & VISUALIZATION

YOUR DATA SOURCE

From Track 1: Cleaned StatsCan Vancouver prices CSV (Week 1)

MAPPING TOOLS

Google My Maps (Recommended - No Coding)

- **Tutorial:** <https://www.google.com/earth/outreach/learn/visualize-your-data-on-a-custom-map-using-google-my-maps/>
- **Free:** Yes
- **Features:** Import CSV, color-code pins, add pop-ups, share link
- **Mobile:** Automatically responsive

Leaflet.js (Advanced - Requires Coding)

- **Tutorial:** <https://leafletjs.com/examples/quick-start/>
- **Free:** Open source
- **Features:** Full customization, can pull from Google Sheets API
- **Requires:** HTML/CSS/JavaScript knowledge

INFOGRAPHIC TOOLS

Canva

- **Link:** <https://www.canva.com> (free tier)
- **Templates:** Search "data visualization" or "infographic"
- **Export:** PNG for social media (1080x1080 Instagram, 1200x628 Facebook)

Datawrapper

- **Link:** <https://www.datawrapper.de> (free)
- **Best for:** Line charts, bar charts showing price trends
- **Export:** PNG or embed code

Flourish

- **Link:** <https://flourish.studio> (free for public projects)
- **Best for:** Animated visualizations

REAL-WORLD EXAMPLES

USDA Food Access Research Atlas

- <https://www.ers.usda.gov/data-products/food-access-research-atlas>
- **Lesson:** Regional averages are valuable (you don't need individual store prices)

ILSR Grocery Map

- <https://ilsr.org/article/independent-business/grocerygaps/>
- **Lesson:** Color-coding by store type, simple filters

COLOR ACCESSIBILITY

Must test for color blindness:

- **Simulator:** <https://www.color-blindness.com/coblis-color-blindness-simulator/>
- **Safe combos:** Dark green + dark blue + orange (NOT red/green together)
- **High contrast:** Always test on mobile screen

DELIVERABLE

Live Google My Maps link with 20-30 stores, 5-7 PNG infographics, one-page brand style guide



TRACK 4: CONTENT & COMMUNICATIONS

Your Data Source

From Tracks 1 & 2: Analysis results, affordability rankings, savings data

Writing Tools

Google Docs (Recommended)

- Free, collaborative
- Easy commenting and version control
- Share with 'anyone can comment'

Grammarly

Website: <https://www.grammarly.com>

- Free tier available
- Checks grammar, clarity, tone

Hemingway Editor

Website: <https://hemingwayapp.com>

- Free web version
- Makes writing clear and simple
- Highlights complex sentences

Plain Language Resources

Plain Language Guidelines (Canada): <https://www.canada.ca/en/treasury-board-secretariat/services/government-communications/canada-content-style-guide.html>

Official guide for clear government writing

Writing for the Web: <https://www.nngroup.com/articles/how-users-read-on-the-web/>

Best practices for online content

Community Guides - What to Create

Student Savings Guide

- Best affordable stores near universities
- Budget meal planning tips
- Recommended items for small households

Newcomer Guide

- Introduction to Metro Vancouver grocery landscape
- Store types explained (discount vs. premium)
- How to compare prices

Family Budget Guide

- Savings strategies for families of 4+
- Bulk buying tips
- How to use the affordability map

Senior/Fixed Income Guide

- Stores with senior discounts
- Transit-accessible options
- Small-portion shopping tips

Neighborhood Guides

- Create 3-5 guides for specific neighborhoods
- 'Where to shop in [neighborhood]'
- Include transit info

Social Media Content

Platform Specs

- Instagram: 1080x1080 (square posts), 1080x1920 (stories)
- Facebook: 1200x630 (link posts)
- Twitter/X: 1200x675
- LinkedIn: 1200x627

Content Ideas

- 'Did you know?' facts about food prices
- Before/after savings examples
- Store spotlight posts
- Map launch announcement
- Weekly tips series

Storytelling Framework

Case Studies Should Include:

- Real (or composite) person's situation
- Their food affordability challenge
- How they used the map
- Actual savings achieved
- Emotional impact ('Now I can afford...')

Example Structure:

- 1. Meet [Name] - their situation
- 2. The Challenge - specific problem
- 3. Discovery - how they found the map
- 4. Results - concrete savings
- 5. Impact - what the savings enabled

Report Writing

Final Report Should Cover:

- 1. Executive Summary (1 page)
- 2. Background & Context (food insecurity stats)
- 3. Methodology (how you collected and analyzed data)
- 4. Key Findings (affordability rankings, savings potential)
- 5. Geographic Analysis (neighborhood comparison)
- 6. Community Impact (who benefits most)
- 7. Recommendations (next steps, future improvements)
- 8. Appendices (data sources, methodology details)

Length: 8-12 pages

Tone: Professional but accessible

Audience: Community organizations, funders, media

Your Deliverable

Final report (clear and accessible), 3–5 short community guides, social media content (posts + key messages), 2–3 short stories or case studies

UNIVERSAL RESOURCES (All Tracks)

Project Management:

- **Trello:** <https://trello.com> (task boards)
- **Asana:** <https://asana.com> (project timelines)
- **Discord:** <https://discord.com> (team communication)
- **Slack:** <https://slack.com> (team communication)

File Sharing:

- **Google Drive:** Unlimited storage @glocalcanada.onmicrosoft.com
- **GitHub:** <https://github.com> (for code, version control)

Video Calls:

- **Zoom:** <https://zoom.us> (team meetings)
- **Google Meet:** <https://meet.google.com>

Documentation:

- **Google Docs:** Collaborative writing
- **Notion:** <https://www.notion.so> (wiki-style documentation)

LEARNING RESOURCES

Data Analysis:

- **Khan Academy Statistics:** <https://www.khanacademy.org/math/statistics-probability>
- **Google Sheets Tutorial:** <https://support.google.com/docs/answer/6000292>

Design:

- **Canva Design School:** <https://www.canva.com/learn/>
- **Color Theory:** <https://www.interaction-design.org/literature/article/the-basics-of-color-theory>

Writing:

- **Purdue OWL:** https://owl.purdue.edu/owl/general_writing/index.html
- **Plain Language Guide:** <https://www.plainlanguage.gov/guidelines/>

Web Development (Track 3 advanced):

- **FreeCodeCamp:** <https://www.freecodecamp.org>
- **MDN Web Docs:** <https://developer.mozilla.org>

SUCCESS CHECKLIST

All Tracks

- ☐ Understand project goals and your role
- ☐ Connected with other tracks for coordination
- ☐ Know which tools you're using
- ☐ Set up collaboration platforms (Discord/Slack)
- ☐

Track 1: Data Collection & Research

- ☐ StatsCan data downloaded and cleaned
- ☐ Master database structure created
- ☐ 50+ stores documented

Track 2: Research & Data Analysis

- ☐ Affordability methodology defined
- ☐ Savings calculator built
- ☐ Analysis charts created

Track 3: Design & Visualization

- ☐ Map built and user-tested
- ☐ Visual branding established
- ☐ Infographics created

Track 4: Content & Communications

- ☐ Final report drafted
- ☐ 3-5 community guides written
- ☐ Social media content calendar created

TIPS FOR SUCCESS

Track 1 (Data)

- ☐ Start with StatsCan - it's reliable and free
- ☐ Document sources, dates, methods
- ☐ Quality over quantity

Track 2 (Analysis)

- ☐ Keep formulas simple and commented
- ☐ Create charts that tell a story
- ☐ Always cite sources

Track 3 (Design)

- ☐ Mobile-first design
- ☐ Test with real people
- ☐ High contrast colors

Track 4 (Content)

- ☐ Write for 8th grade reading level
- ☐ Use short sentences and paragraphs
- ☐ Include real examples

All Tracks

- ☐ Communicate early and often
- ☐ Ask for help when stuck
- ☐ Document your work for future volunteers